

Gaussian Processes

- Gaussian Processes (GPs) allow us to make predictions about data by incorporating prior knowledge. For a given set of training points, there are potentially infinitely many functions that fit the data. GPs offer an elegant solution by assigning a probability to each of these functions. The mean of this probability distribution represents the most probable characterization of the data. Using a probabilistic approach allows us to incorporate the confidence of the prediction into the result.
- A random variable $\mathbf{x} \in \mathbb{R}^n$ follows a multivariate gaussian dist in n dimensions with mean $\mu \in \mathbb{R}^n$ and covariance $\Sigma \in \mathbb{R}^{n \times n}$ if:

$$p(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$$
- A GP is an extension of the multivariate gaussian to infinite dimensions. Given an input vector $x \in \mathbb{R}^n$, the GP returns a transformed vector $y \in \mathbb{R}^n$. Every component of y represents the *probability* of observing x_i according to some Gaussian living in dimension i .
- Since a GP acts as a probability distribution, we can also sample new y for new x apart from transforming an x into a y . Hence, if we sample the entire domain of x , we can obtain the function f such that $f(x) = y$.

Marginalization: Integrate along one of the dimensions of the Gaussian to get the probability distribution over the remaining dimensions: $P(X) = \int_{-\infty}^{\infty} P(X, Y) dY$ or $P(X) = \Sigma_Y P(X, Y)$.

Conditioning: Updates the distribution over function values at new points, given observations at known points. By conditioning on observed data, GPs allow us to effectively incorporate priors and make informed predictions in uncertain scenarios:

$$X|Y \sim \mathcal{N}(\mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (Y - \mu_Y), \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX}) \quad Y|X \sim \mathcal{N}(\mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (X - \mu_X), \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY})$$

^^ Complex math, but note that the updated means only depend on the conditioned variable, while the covariance matrix is independent of the variable.

Functional view of GPs

- GPs (which are stochastic processes) essentially are a set of random variables. Let each of these random variables have a corresponding index i , which represents the i^{th} dimension of the n -dimensional multivariate distribution.
- Goal: Learn the underlying distribution from *train data* (Y).
- GPs attempt to model the underlying distribution of the *test data* (X), by modeling the joint distribution $P_{X,Y}$ - the span of possible function values - as a multivariate normal distribution whose dimensionality is $|X| + |Y|$.
- Regressing on X requires modeling this as Bayesian inference, i.e., update the (current) hypothesis as information (Y) becomes available. This conditional probability ($P_{X|Y}$) is a multivariate normal distribution as well.

- GPs treat each test point as a random variable. Making predictions essentially requires sampling from the distribution; the i^{th} component of the sampled vector is interpreted as the function value corresponding to the i^{th} test point.

Setting up the distribution requires defining μ and Σ . In GPs, centering the data, i.e., $\mu = 0$ is common practice. The covariance matrix (Σ) is determined by the GP's covariance function k - the kernel of the GP.

Kernels

- The covariance matrix describes the shape of the distribution and determines the characteristics of the predicted function.
- Σ is determined by evaluating the kernel/ covariance function (k) pairwise on all the points and acts as a similarity measure: $k : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$; $\Sigma = \text{Cov}(X, X') = k t, t'$
- Σ_{ij} describes how much influence the i^{th} and the j^{th} point have on each other. Since k describes the pairwise similarity between the function values, it controls the possible shape that a fitted function can adopt. This formulation allows the function to take on similarity measures beyond simple Euclidean distances.
- Stationary kernels such as the RBF kernel ($\sigma^2 \exp(-\frac{\|t-t'\|^2}{2l^2})$) or the periodic kernel ($\sigma^2 \exp(-\frac{2 \sin^2(\pi|t-t'|/p)}{l^2})$) are invariant to translations and the covariance of two points is only dependent on their relative position. Non-stationary kernels such as the linear kernel ($\sigma_b^2 + \sigma^2(t-c)(t-c')$) do not have this constraint and depend on an absolute location.
- Different kernels can describe different classes of functions, which can be used to model the desired shape of the function.

Combining Kernels

- Kernels can be combined together resulting in a more specialized kernel, allowing domain experts to introduce necessary priors to capture trends in the data.
- The covariance matrix of GPs has to be positive semi-definite. When choosing optimal kernel combinations, all methods that preserve this property are allowed (such as addition and multiplication).

Prior Distribution

- Training data is the *prior* distribution (P_Y). Since centering the distribution ($\mu = 0$) is the general convention, we begin with $\mu = 0$ with dimensionality $|Y| = N$.

- The kernel function (dimensionality $N \times N$) determines which functions from the space of all possible functions are more probable.
- Adjusting the kernel function parameters controls the shape of the resulting functions and also varies the confidence of the prediction. For example, decreasing the variance (σ) results in sampled functions that are concentrated around the mean (μ).

Posterior Distribution

- Model the joint distribution ($P_{X,Y} \in \mathbb{R}^{|Y|+|X|} \times \mathbb{R}^{|Y|+|X|}$) and use train data (Y) to constrain the distribution and infer test data (X), i.e., the output is the conditional probability $P_{X|Y} \in \mathbb{R}^{|X|}$.
- Using conditioning, $P_{X|Y}$ can be derived from $P_{X,Y}$. Conditioning leads to derived versions of μ and Σ : $X|Y \sim \mathcal{N}(\mu', \Sigma')$ where $\mu' \neq 0$. This is because the training datapoints (Y) constrain the set of possible functions to those that pass through the training points.
- Further, precisely passing through Y may lead to overfitting/ over-complexity, especially with noisy real-world data. We model noise/ measurement errors with the error term $\epsilon \sim \mathcal{N}(0, \psi^2)$: $X = f(Y) + \epsilon$.
- The resulting joint probability distribution is:

$$P_{X,Y} = \begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}(0, \Sigma) = \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} + \psi^2 I \end{bmatrix}\right).$$
- We obtain a prediction for our function values by sampling from the conditional distribution which is derived from conditioning on the joint distribution described above.
- Since sampling involves randomness, predictions may not be deterministic. Marginalizing (the noise) associated with each random variable (i) allows us to extract the function value (μ'_i) and uncertainty ($\sigma'_i = \Sigma'_{ii}$).

Take II: Gaussian Processes; not for dummies.

(These notes are based on the notes taken while summarizing a talk $\implies$ It does not lend itself to compact note taking. However, the visual explanations in the [blog post](#) are really helpful for building intuitions.)

Motivation: Non-Linear Regression

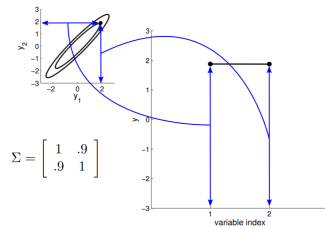
Given a few (training) datapoints $\{X_i, Y_i\}$, we try to estimate Y_j at a yet unobserved X_j . Fitting a non-linear curve to $\{X_i, Y_i\}$ is generally the solution. This results in a single function that is considered the best fit given $\{X_i, Y_i\}$ without accounting for the possibility that future observations may not resemble the current hypothesis, i.e., there is no uncertainty associated with the current estimate.

The World of Gaussians

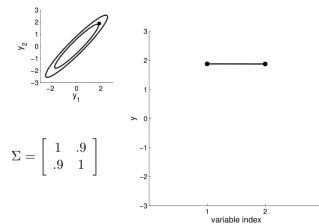
- A random variable is k -variate normally distributed if every linear combination of its k components has a univariate normal distribution: $X = (X_1, \dots, X_k)^T$ is a multivariate Gaussian distribution if $Y = a_1X_1 + a_2X_2 + \dots + a_kX_k$ is normally distributed for any constant vector $a \in \mathcal{R}^k$.

2D Gaussians

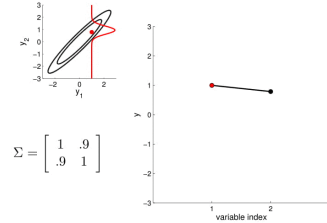
- New notation for sampling from 2D Gaussians: Take the oval contour graph (top-left). Choose a random point on the graph. Plot the value of y_1 and of y_2 that point on graph at index=1 and index=2:



- The sampling operation becomes a mapping of 2D points from the oval contours (top-left) to plotting at index=1 and index=2. Since y_1 and y_2 are correlated, index graph samples can only move away from each other within some margin:

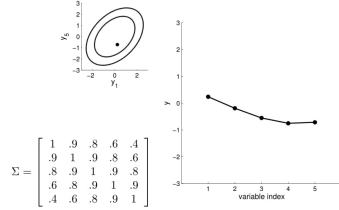


- Conditioning: Simply fix one of the endpoint on the index graph (say, fix y_1 to 1) and sample from y_2 :

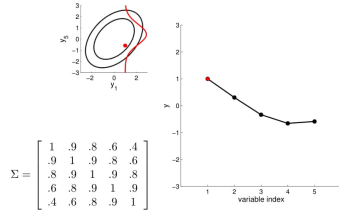


5D Gaussians

- Index Sampling: The 2D case discussed above extends to 5D Gaussians:



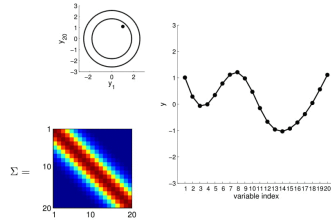
- Conditioning: The 2D case discussed above extends to 5D Gaussians :



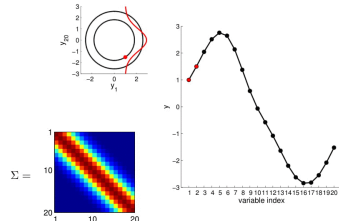
- Non-linear regression as curve fitting is now starting to emerge from index sampling. The correlations from the covariance matrix also constrain how the index samples are able to express themselves.

20D Gaussians

- Index Sampling:

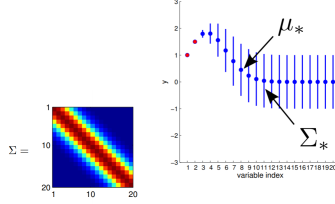


- Conditioning:



- Non-linear regression reveals itself in the index plot. It is possible to generate a family of curves that fits the observations with one sample from the Gaussian. If we sample multiple times, we can compute the

mean and variance of the fit:



Gaussian Processes

- A GP is a collection of random variables, any finite number of which have consistent Gaussian distributions. The GP is completely defined by (1) a mean fncn $m(x)$ and (2) a covariance fncn $K(x, x')$. The mean can be any value and the covariance matrix should be PSD: $f(x) \sim \mathcal{GP}(m(x), K(x, x'))$
- GPs are non-parametric while non-linear regression is parametric. But, a non-parametric model is simply a parametric model with ∞ parameters.
- A parametric model explicitly defines its parameters: $y(x) = f(x) + \epsilon\sigma_y$, where $p(\epsilon) = \mathcal{N}(0, 1)$ and σ_y is Gaussian noise indicating how noisy the fit is to the actual observation.
- A GP prior can be placed over the above function: $p(f(x)|\theta) = \mathcal{GP}(0, K(x, x'))$, where $K(x, x') = \sigma^2 \exp(-\frac{1}{2l^2}(x - x')^2)$.
 - σ (vertical scale): Describes how much span the function has, i.e., the overall function variance/ amplitude. A higher σ allows for larger variations in function values even for nearby points, whereas a smaller σ restricts the function to vary within a narrower range.
 - l (horizontal scale): Describes how quickly the correlation between two points drops as their distance increases. A high l gives a smooth function. A low l results in a wiggly function - points become uncorrelated over short distances allowing for rapid changes.
 - Since $p(y|\theta)$ is Gaussian, we can compute the likelihood in close form. We simply maximize the likelihood of $p(y|\theta)$ under the hyperparameters using a gradient optimizer: $\arg \max_{l, \sigma^2} \log p(y|\theta)$.
- The above GP can also incorporate Gaussian noise (σ_y) directly into the model (sum of Gaussians is a Gaussian): $p(f(x)|\theta) = \mathcal{GP}(0, K(x, x') + I\sigma_y^2)$

References

1. [Gaussian Processes](#) - Reid Pryzant.
2. [A Visual Exploration of Gaussian Processes](#) - Jochen Görtler, Rebecca Kehlbeck, and Oliver Deussen.
3. [Gaussian Processes, not for dummies](#) - Yuge Shi.